

Announcements

- Next week is Fall Break - No School
- This is the first semester this lab has been taught since it was updated, so any feedback is appreciated
- Reminder: You must be present to receive credit for labs
 - If you can't attend a given week, please let me know in advance so I can figure out a way for you to makeup the lab
- Additionally, please **do not** have AI do your work for you

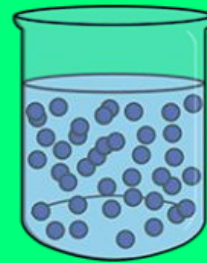
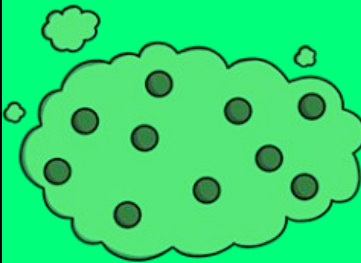
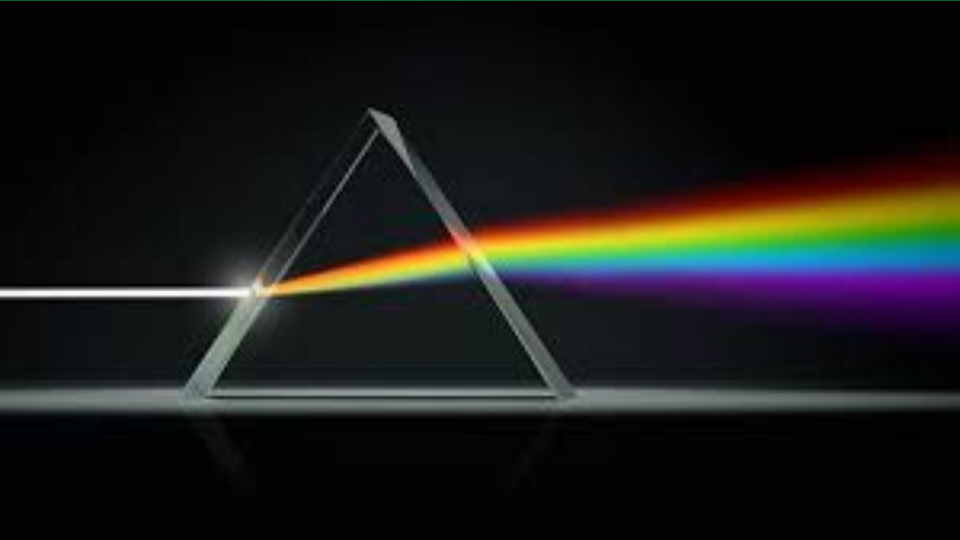
Announcements

- Next week is also a day lab, so meeting in **LSE-S04** at **3:30 p.m.**
- This is the first semester this lab has been taught since it was updated, so any feedback is appreciated
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Spectroscopy

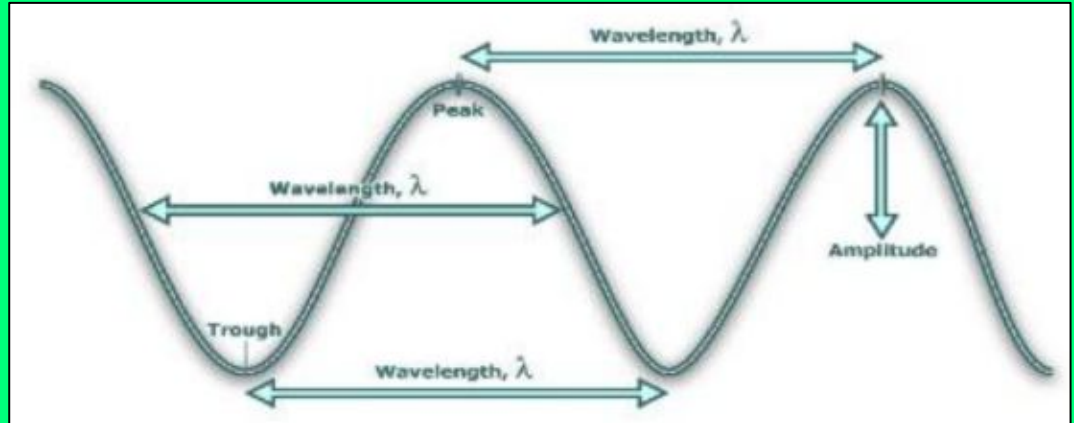
What is Spectroscopy?

- Studying the interactions between light and matter



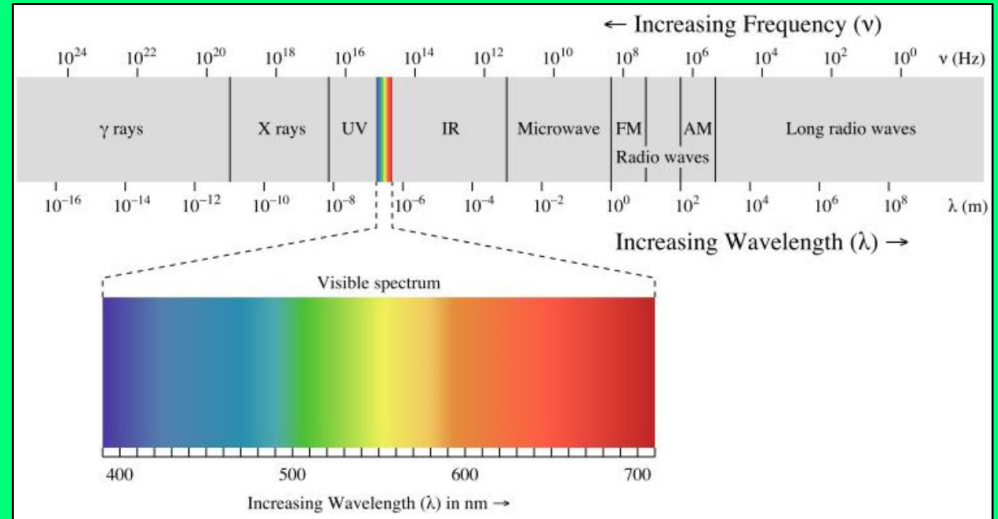
Light

- It's a wave
- For this lab, you want to pay the most attention to wavelength (λ)
 - The distance between two peaks or two troughs
 - The inverse is frequency (f)
 - units of Hertz (per second)
- c (the speed of light) = $\lambda * f$ (distance per time)
- Amplitude is how “tall” the wave is



Light is a Spectrum

- We divide it into different sections based on wavelength
- You can think of it as different wavelengths of light being different colors
- Shorter wavelengths (higher frequencies) are bluer
 - Longer wavelengths are redder
 - Shorter wavelengths are also higher energy - think about it as being hit by more waves per second



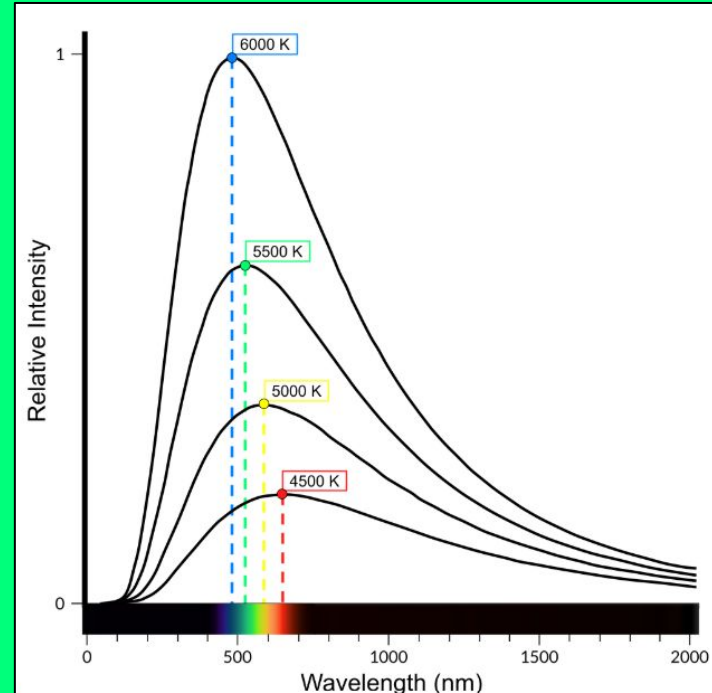
Wien's Law

- Everything emits light based on the temperature (“black-body radiation”)
 - higher temperature = higher energy = shorter wavelength = bluer
- If you plot out all the wavelengths emitted, you get a curve like these (called a Planck curve) →
- The peak of that curve is the wavelength where the most light is emitted
- This wavelength can be calculated using Wien’s Law:

$$\lambda_{peak} = \frac{2.9 \times 10^6}{T}$$

cough questions 20 and 21 *cough*

Any guesses on why the background is so aggressively green?



Spectra (Plural of “Spectrum”)

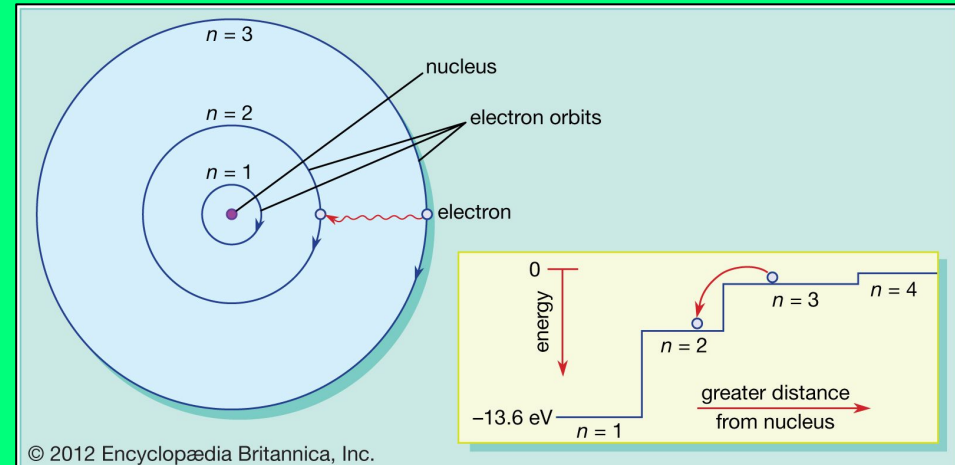
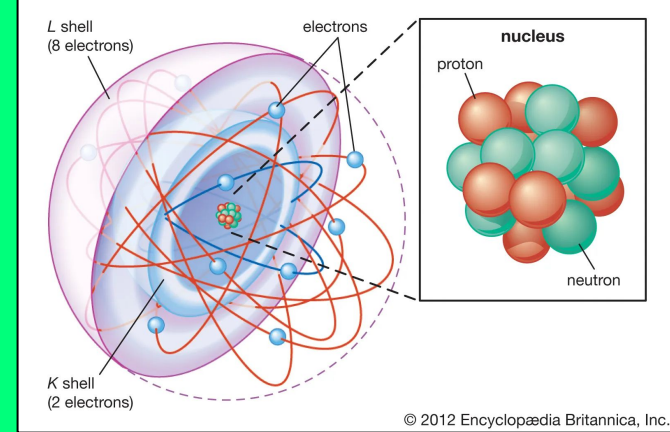


- You can use a prism to separate the light into its constituent wavelengths (colors)
 - A natural version is a rainbow - where the prism is water droplets suspended in the air
- This happens because different wavelengths refract (deflect or bend) by different amounts
- We don't always see a full spectrum when we observe things
 - Sometimes only a few wavelengths (“colors”) are there
 - Sometimes it's mostly there, but a few are missing
- We need to look at how light interacts with matter to figure out why we don't see the full spectrum
 - Note that this is different from the black-body radiation of Wien's law



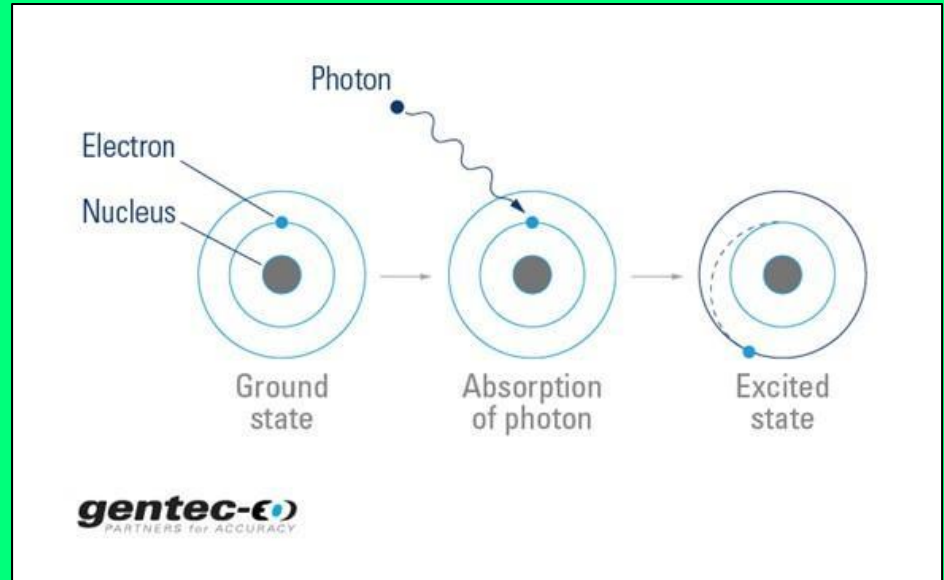
A Brief Aside About Atoms

- A nucleus made of protons and neutrons
 - Not too important for today's lab
- Surrounded by electrons
- The electrons can be separated into different “shells” or “orbits”
- Each shell represents a single quantized energy level, increasing as you get further from the nucleus
 - The word “quantized” here is important - it means a specific, discrete (not a continuum) amount
 - Also where the word “quantum” as in quantum mechanics comes from
 - Electrons orbiting an atom can **only** exist at these specific energy levels
 - Each shell has a maximum number of electrons it can hold - add more, they go to the next shell



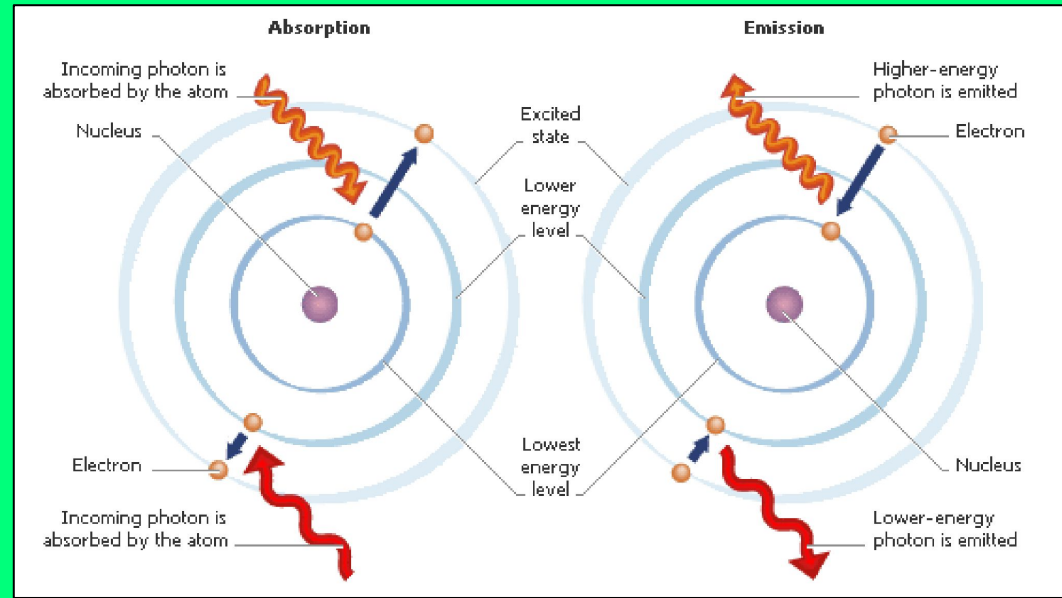
What Happens When Light Hits an Atom?

- Absorption!
- But light has energy
- And energy can never be created or destroyed*
 - *(Outside nuclear reactions)**
 - **(But matter is equivalent to energy, see $E = mc^2$)
- So the total energy has to be the same. Where did it go?
 - One of the electrons took that energy and went up to a higher energy level
 - Sometimes, the energy is so much that the electron leaves the atom altogether - this process is called “**ionization**” and the atom is said to be “**ionized**”

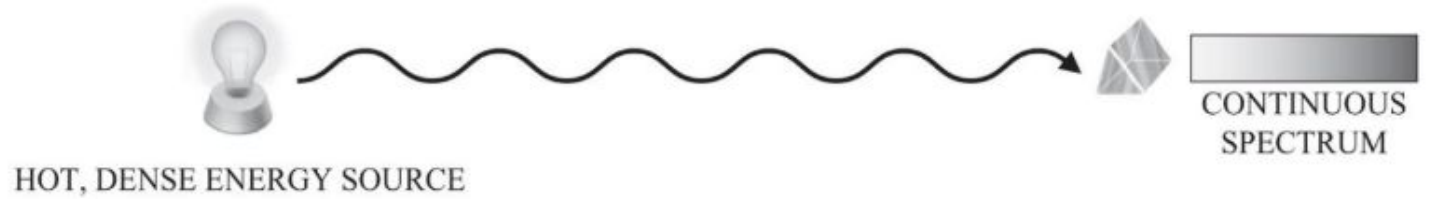


But it Doesn't Stay There

- Electrons don't like having a lot of energy, they want to be as close to the "ground state" as they can
- When an electron drops back down to a lower energy level, the energy it did have still has to go somewhere
- So, the atom emits light
- The quantized amount of energy absorbed or emitted depends on the atomic structure of the atom - so atoms absorb and emit different **specific** wavelengths
 - The more levels up or down - the bigger the energy change
- So by looking at what colors are absorbed or emitted; we can determine the composition

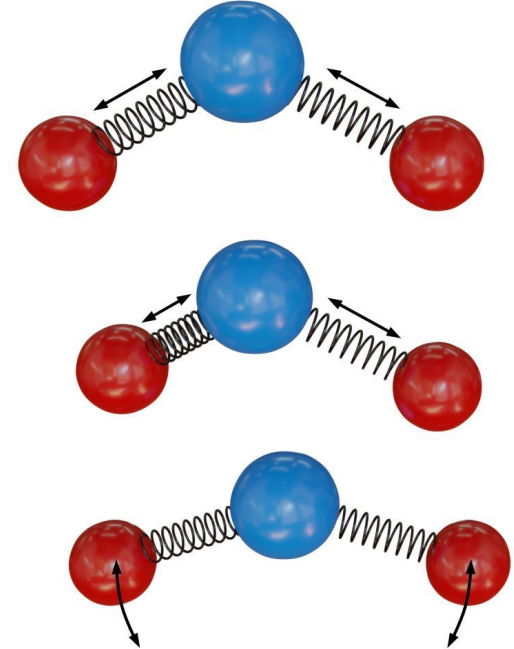
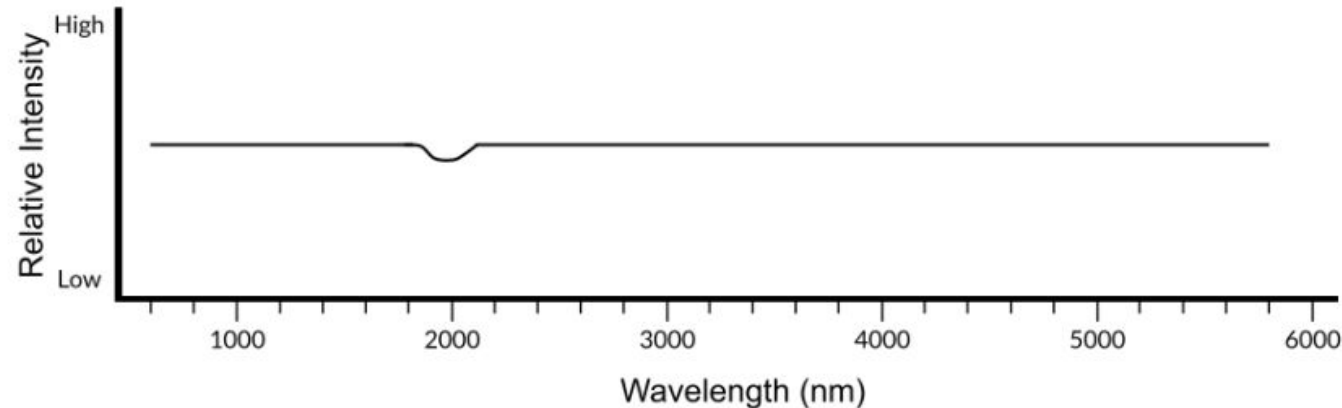


How This Looks

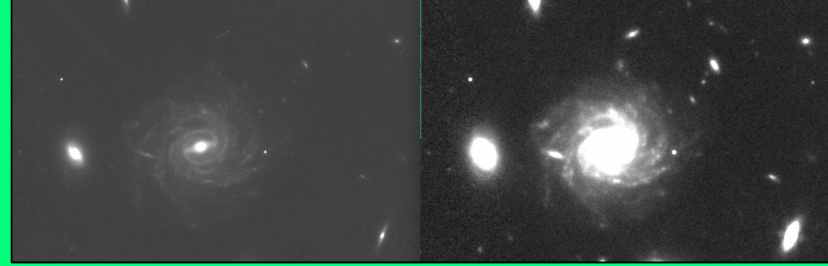


Molecular Spectroscopy

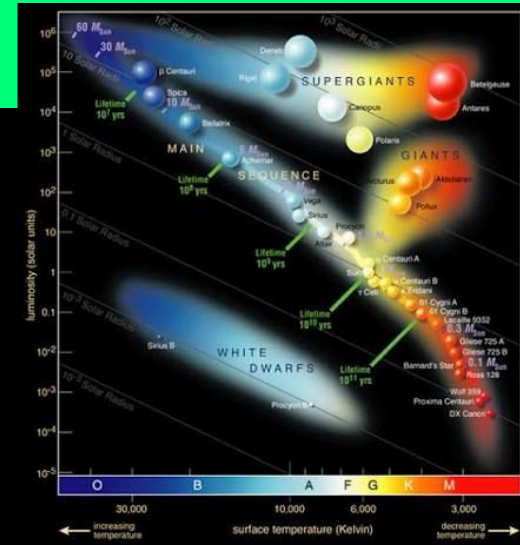
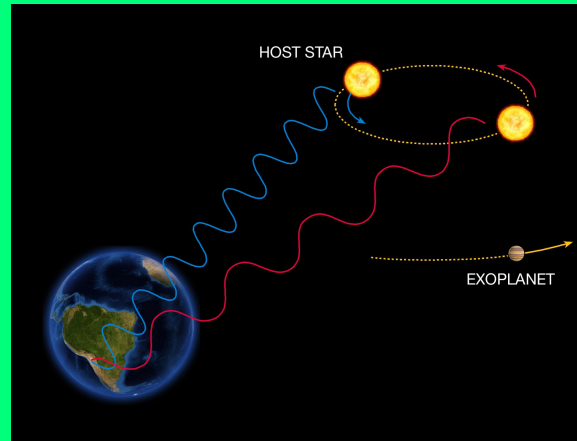
- Molecules, being made of multiple atoms, make things a lot more complicated
 - The bonds between atoms move in different ways
 - Vibrating and bending
- This means when looking at the spectra, you see much broader features, rather than clear lines



In Astronomy

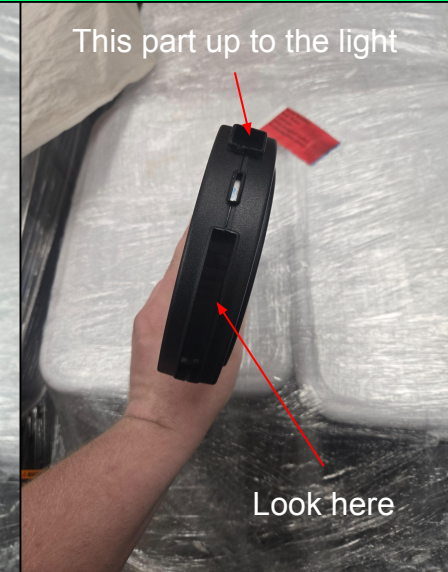
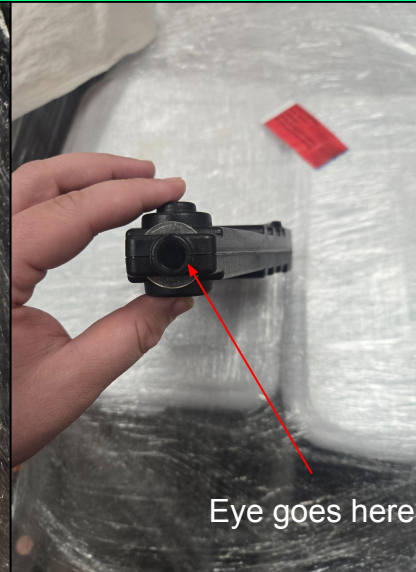


- Spectroscopy can tell us a lot about things that are far away - which is most of what we study in astronomy
- Exoplanets (radial velocity and atmospheric spectroscopy)
 - Later this semester
- Star temperatures (and therefore mass and age - H-R Diagram)
 - Learn in more detail in AST 112
- Composition of gas clouds, comets (what you are doing today), and other astronomical objects



How to use a Spectrograph

- The light goes in the square hole
- Put the square hole up to the light source
 - As close as you can
- Look straight, i.e. at the rectangular part



Today's Lab

- Work through questions 1-12
- Use the spectrosopes to look at each of the 6 elements we have lit up
 - Identify which is which and sketch the emission lines on your worksheet
 - Please return the colored pencils in the same section of the box when you are done
 - You can view the spectroscopy stations in any order, but the answers do need to be in order
 - Answer questions 14-18 based on your results
- Work through the rest of the lab questions

Image Credits (In order of appearance)

- Twinkl
 - <https://www.twinkl.com/parenting-wiki/states-of-matter>
- Design Basics
 - <https://designbasics.forumist.com/color/>
- Old version of the lab script
- Old version of the lab script
- National Geographic
 - <https://education.nationalgeographic.org/resource/rainbow/>
- The lab script
- Encyclopedia Britannica - Structure of the nucleus
 - <https://www.britannica.com/science/atom/Structure-of-the-nucleus>
- Gentec EO
 - <https://www.gentec-eo.com/blog/what-is-stimulated-emission-in-lasers>

Image Credits (continued)

- Old version of the lab script
- Tec Science
 - <https://www.tec-science.com/thermodynamics/kinetic-theory-of-gases/equipartition-theorem/>
- The lab script “Mission Training”
- Encyclopedia Britannica - Atom
 - <https://www.britannica.com/science/atom>
- JWST image, in f090w and f444w filters
- ESA
 - <https://www.cosmos.esa.int/web/cesar/the-hertzsprung-russell-diagram>
- ESO
 - <https://www.eso.org/public/images/eso0722e/>
- Taken by me (X3)